

Project Design Document

Team Number: 10, Project 1

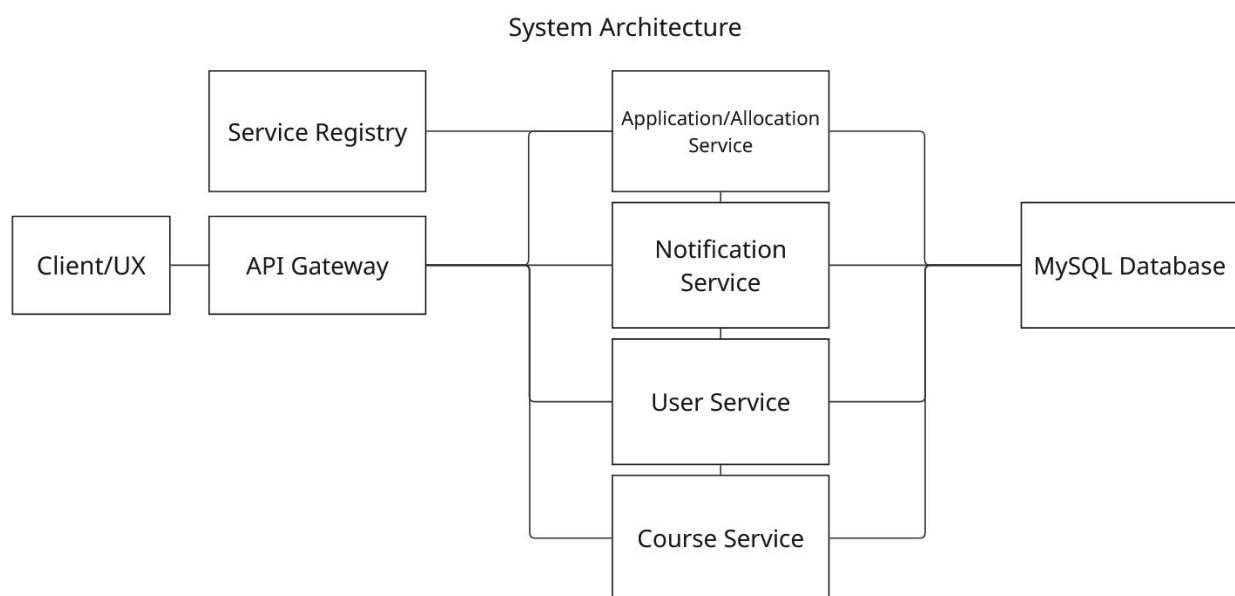
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Introduction

We are building a TA application portal that will function to provide a more organized and connected system for TA management. TA's will be able to create accounts and easily apply for TAship's in upcoming semesters. Instructors will be able to use the system to input what they require for courses including grading and course preferences, as well as visualize the current schedule and TA allocation. The main actor in the system will be the TA coordinator, who will be able to visualize student schedules and course assignments, view the instructor needs and student preferences, and try to match appropriately. They will also be able to upload a CSV of the upcoming course schedule into the system. This will allow them to send offers to TA's, and overall manage the system and its users.

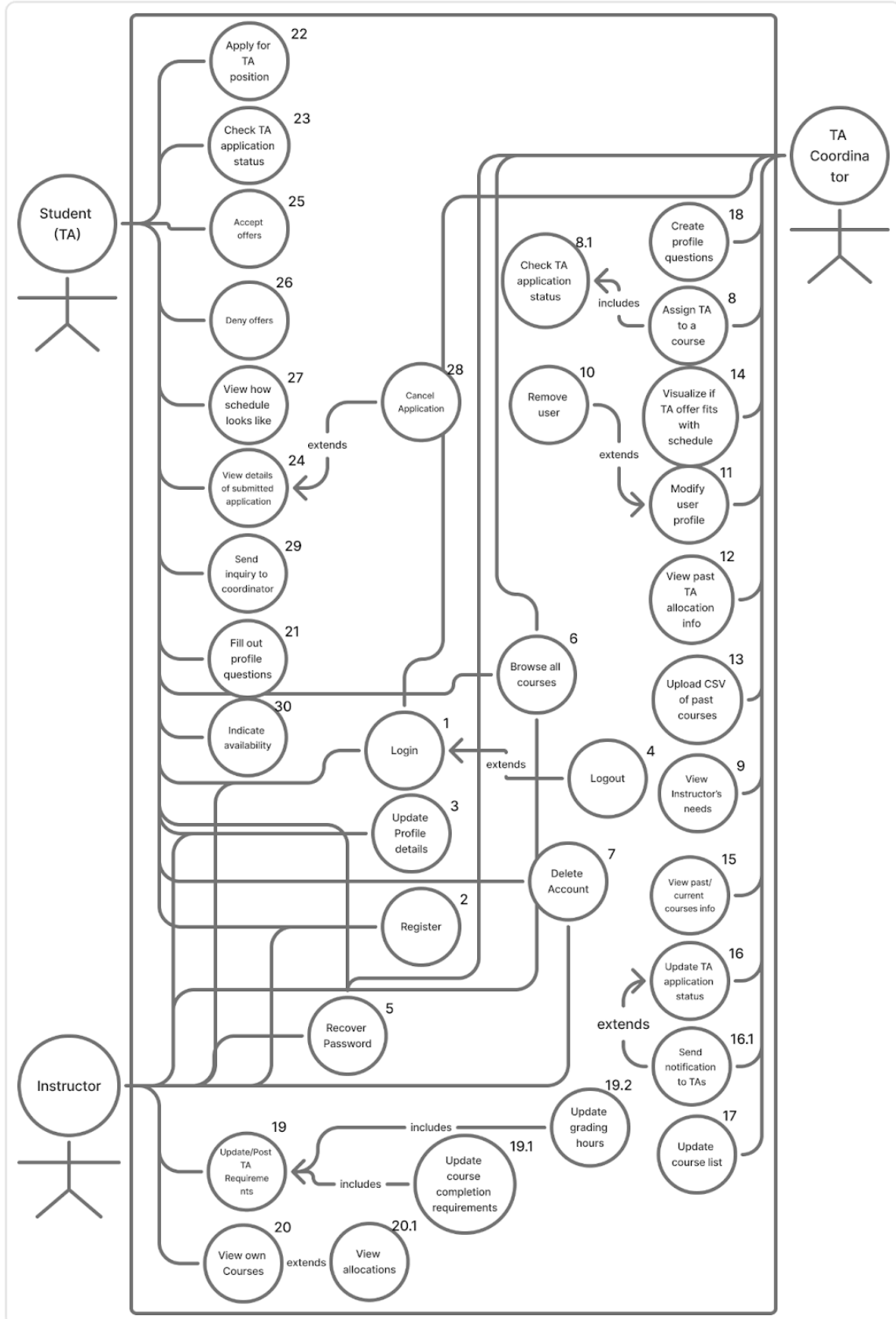
System Architecture



The frontend client will make requests to the API Gateway. The API Gateway and all the other services are registered in the service registry. When the user calls something, the API Gateway will know about their existence through the service registry, and will then put their API request to the appropriate service. The services then all can communicate between each other for any data they need, and they store all their own information in their own database inside the MySQL container.

Use Case Models

Use Case Diagram



Use Case 1: Login

Primary Actor: Student, Instructor, TA-coordinator

Description: Describes the process of how a user logs in

Precondition: The user is logged in

Postcondition: The user logs in

Main Scenario:

1. The user enters their email address and password to login.
2. The user logs in.

Extensions:

Use Case 2: Register

Primary Actor: Student, Instructor

Description: Describes the process of how a user can register as a student or instructor.

Precondition: The user wishes to register to the site.

Postcondition: The user has registered and can login to the site.

Main Scenario:

1. The user is in the login page and clicks on "Register account"
2. The user registers their account, including whether they are a student or instructor.
3. The application verifies the registration by sending an email to the user.
4. The user verifies the registration after clicking on the link sent by the application.
5. The user's account is not enabled and can now login.

Extensions:

1. The user finds that their email address is taken for some mysterious reason.
 - The user can view in the footer a way to contact the admin of the site, which is the TA-coordinator.
2. The user accidentally writes a typo in their name or other important information
 - The user can go do Use Case: Update Profile Details. Most likely, he or she will have to contact the TA-coordinator (admin).

Use Case 3: Update profile details

Primary Actor: Student, Instructor

Description: Describes the process of how a user can update profile details.

Precondition: The user is logged in and wishes to update their profile.

Postcondition: The user has updated their profile.

Main Scenario:

1. The user logs in and clicks on their account details.
2. The user clicks on a button to change their account details.
3. The user changes their email, or other information. Their name and student number may be uneditable. When changing the email, the user needs to undergo the verification process as in the Register Use case.
4. The user submits and the profile is updated.

Extensions:

1. The student wishes to change their name and student number.
 - The user must contact the TA-coordinator.
2. The instructor wishes to change their name and course department and other possible sensitive information.
 - The user must contact the TA-coordinator.

Use Case 4: Logout

Primary Actor: Student, Instructor, TA-coordinator

Description: Describes the process of how a user logs out.

Precondition: The user is logged in.

Postcondition: The user is logged out.

Main Scenario:

1. The user is logged in and clicks on log out.
2. The user is logged out.

Extensions:

Use Case 5: Recover Password

Primary Actor: Student, Instructor, TA coordinator

Description: Describes the process of how a student, instructor, or TA coordinator can recover their password

Precondition: The user wishes to login.

Postcondition: The user has recovered their password and can successfully login.

Main Scenario:

1. The user is in the login page and forgets his or her password. The user clicks on "Recover password"
2. The user is sent to a new Recover Password page and the user types out their email address. The application sends an email with a code.
3. The user copies the code from the email and types it out on the website. The application sends an email with a link that enables them to change their password.
4. The user changes their password after clicking the link, and can now login.

Extensions:

1. The user thinks they registered to the site but actually didn't. The user inputted their email address into the "Recover Password" page.
 - the application informs the user that the email address doesn't exist.

Use Case 6: Browse all courses

Primary Actor: Student, Instructor, TA coordinator

Description: Describes the process of how registered users can browse all courses.

Precondition: The User must have registered and is a UBC student, instructor, or TA coordinator.

Postcondition: The user has viewed all the courses related to TA allocation, which the coordinator has inputted.

Main Scenario:

1. The user logs in and clicks on browse courses.
2. The user can browse the courses by searching and/or filtering

Extensions:

1. The user searches with the wrong keyword
 - the application does not return anything.
2. The user does not find a course that should be there
 - the coordinator can be contacted and the coordinator can update the list of courses.

Use Case 7: Delete Account

Primary Actor: Student, Instructor

Description: Describes the process of how a student or instructor can delete their account.

Precondition: The User must have registered and is a UBC student or instructor.

Postcondition: The user has deleted their account and can no longer login to the site.

Main Scenario:

1. The user logs in and clicks on my account.
2. The user clicks on "delete account"
3. The application verifies the deletion. All related information to the user such as applications are deleted from the database.
4. The user is logged out.

Extensions:

1. The user accidentally clicks on delete account
 - The application verifies by asking the user to type out something specific, like their email address. Only if this is satisfied does the application ask one more time if the account can be deleted.

Use Case 8: Assign TA to a course

Primary Actor: TA Coordinator

Description: Describes the process when a TA coordinator assigns a TA to a course.

Precondition: TA coordinator must have been successfully logged in and TA must have submitted a TA application.

Postcondition: If assigning TA to a course is successful, TA course-related databases are updated.

Main Scenario:

1. The TA coordinator does Use Case 14: Visualize if TA offer fits with schedule.
2. The TA coordinator clicks on the button to allocate the TA.
3. The TA coordinator is prompted to verify the course assigned.
4. If assigning is successful, database entry corresponding to TA is updated and TA application marked as completed.
5. The email notification of the allocation to the TA is not sent yet and neither is the application status updated yet.

Extensions:

1. If TA application is submitted incorrectly with missing details
 - TA coordinator can deny the application and email the TA to apply again.

Use Case 9: View instructors needs

Primary Actor: TA Coordinator

Description: Describes the process when a TA coordinator is viewing instructors' needs for the TA.

Precondition: TA coordinator must have been successfully logged in and instructor must have posted TA requirements.

Postcondition: If viewing is successful and the instructor has posted TA requirements, instructors' needs are displayed to the TA coordinator.

Main Scenario:

1. The coordinator can click on the button that shows courses that still require TA allocation.
2. The TA coordinator can view instructors' requirements when scrolling through the courses and/or clicking on them.
3. If viewing is successful and the instructor has posted TA requirements, the requirements are shown to the TA coordinator.

Extensions:

1. If requirements incorrectly displayed or not displayed at all

- TA coordinator can request the Instructor to post TA requirements again.

Use Case 11: Modify user profile

Primary Actor: TA Coordinator

Description: Describes the process when a TA coordinator is modifying a user's profile.

Precondition: TA coordinator must have been successfully logged in and user profile must exist.

Postcondition: If modification is successful, user-related databases are updated.

Main Scenario:

1. The coordinator can click on the button to view and search all users.
2. The coordinator selects a user.
3. The TA coordinator clicks on the button to modify a user's profile .
4. The TA coordinator will be able to modify the user's profile details. Additionally, they can remove the user as well (use case: Remove User).
5. The TA coordinator is prompted to verify the updated details.
6. If modifying is successful, database entry corresponding to the TA modified is updated with the new details.

Extensions:

1. If error in modification
 - TA coordinator can do use case:Modify User Profile again to modify users profile.

Use Case 12: View past TA allocation info

Primary Actor: TA Coordinator

Description: Describes the process when a TA coordinator is viewing past TA allocation info.

Precondition: TA coordinator must have been successfully logged in and TA must have past allocation.

Postcondition: If viewing is successful and TA has past allocation, info displayed to TA coordinator.

Main Scenario:

1. The coordinator can click on the button to view and search all users.
2. The TA coordinator selects the TA for whom they want past allocation info.
3. If viewing is successful and TA allocation info for selected TA exists, info displayed to the TA coordinator.

Extensions:

1. If past TA allocation info not displayed
 - TA coordinator can contact TA to request the information.

Use Case 13: Upload CSV of past courses

Primary Actor: TA Coordinator

Description: Describes the process when a TA coordinator is uploading CSV of past courses.

Precondition: TA coordinator must have been successfully logged in and must have the CSV of past courses.

Postcondition: The csv is uploaded and the information is updated in the backend.

Main Scenario:

1. The TA coordinator clicks on the button to upload CSV of past courses.
2. Upload the CSV.
3. The coordinator can manually adjust any data that has been interpreted from the CSV, because the system may have interpreted something incorrectly.
4. If upload and saving is successful, past courses-related database is updated with the information present in the CSV file.
5. The coordinator is shown the information that has been updated.

Extensions:

1. If uploaded file is not in CSV
 - application displays error message and requests for reupload.

2. The system does not read the CSV correctly.
 - the coordinator can manually adjust the data the system interpreted or do Use Case 17: Update Course list to change course information manually.

Use Case 14: Visualize if TA offer fits with schedule

Primary Actor: TA Coordinator

Description: Describes the process when a TA coordinator is visualizing if the TA offer fits with TA's schedule.

Precondition: TA coordinator must have been successfully logged in and TA must have submitted a TA application.

Postcondition: If TA offer fits with schedule, TA course-related databases are updated.

Main Scenario:

1. The TA coordinator clicks on the button to visualize the schedule.
2. The TA coordinator views the scheduler.
3. The coordinator clicks on the TA to put in the schedule and compare with other TAs or instructor's needs of a course.
4. If visualization is successful and the offer fits with TA's schedule, the offer is sent out.

Extensions:

1. If TA schedule is not displayed or displayed incorrectly
 - TA coordinator can view TA's courses and manually update their schedule.

Use Case 15: View past/current courses info

Primary Actor: TA Coordinator

Description: Describes the process when a TA coordinator is viewing past courses info.

Precondition: TA coordinator must have been successfully logged in.

Postcondition: Past/current courses info is displayed.

Main Scenario:

1. The TA coordinator clicks on the button to view all courses.
2. The TA coordinator selects the course for which they want to view past/current info.
3. If viewing is successful, the information of the selected past/current course is shown to the TA coordinator. The coordinator can see the TA allocations of that past/current course.

Extensions:

1. If past courses info displayed incorrectly or not displayed at all
 - TA coordinator can do Use Case: Upload CSV of past courses

Use Case 16: Update Application status of TA

Primary Actor: TA Coordinator

Description: Describes the process when a TA coordinator is marking a course allocation as done

Precondition: TA coordinator must have been successfully finished allocating all or part of the allocation of courses.

Postcondition: Application status of TAs are updated. Some TAs will get an allocation. The TAs may get an email notification depending on the coordinator's decision.

Main Scenario:

1. The TA coordinator will go to view all the users
2. Optionally, the coordinator can filter the users to show all the TAs that have been allocated or have not been allocated.
3. The coordinator can see the TA allocation status when clicking on a user or just from going through the list of users without specifically clicking on the user.
4. The coordinator views their application and can click accept or deny.
5. The system confirms the action and informs the coordinator that the application status of the TAs allocated to those courses will change. A notification will automatically be sent.

Second Scenario:

1. The coordinator goes to the scheduler.
2. The coordinator completes the TA allocations for a course on the scheduler page.
3. The coordinator clicks on a TA without moving from the scheduler page and updates their application status.

4. The system confirms the action and informs the coordinator that the application status of the TAs allocated to those courses will change.

Extensions:

1. The coordinator changed the application status, but wishes to revert it.
 - The coordinator must click on the button to change it back to “deny,” and the system will send an email that the application status has turned into deny.

Use Case 17: Update Course List

Primary Actor: TA Coordinator

Description: Describes the process when a TA coordinator is updating the course list database

Precondition: TA coordinator must have been logged in.

Postcondition: Course list is updated and every other user can see it in the list.

Main Scenario:

1. The coordinator clicks on Courses from the navbar.
2. The coordinator clicks on a button to add a new course.
3. The coordinator fills out a form to add the new course, whether it is a legacy course or a course for a new semester.
4. The course list is updated and everybody else can see it.

Extensions:

1. The coordinator makes a mistake when creating a new course.
 - The coordinator can click on a course and update its details.

Use Case 18: Create profile questions

Primary Actor: TA Coordinator

Description: Describes the process when a TA coordinator creates a questionnaire for the TA-application.

Precondition: TA coordinator must have been logged in.

Postcondition: TA coordinator has saved the questionnaire and the student can see the new questions in his or her application.

Main Scenario:

1. The coordinator clicks on a button to edit the questionnaire.
2. The coordinator fills out or makes changes to the questionnaire.
3. The coordinator saves it and the system renders the old questions deprecated and saves the new questions.

Extensions:

2. The coordinator makes changes or creates the questionnaire as the student is filling out the application
 - The student fills out the application for the now-deprecated questions and the system still retains the information.
 - The coordinator can contact the student to redo the application due to updates to the questionnaire.

Use Case 19: Post/ Update TA requirements

Primary Actor: Instructor

Description: Describes the process when an instructor updates TA requirements.

Precondition: Instructor must have been successfully logged in .

Postcondition: If updating or posting TA requirements is successful, TA requirements-related databases are updated.

Main Scenario:

1. Instructor is on the page for a certain course he is teaching
2. Instructor clicks on the button to update TA requirements.
3. The instructor completes update/post grading hours updating/post course completion requirements.
4. The instructor is prompted to verify the updated details.
5. If the update is successful, database entry corresponding to original requirements is updated.

Extensions:

1. Instructor updated requirements incorrectly
 - Instructor can do use case: Update TA requirements again to reupdate requirements.

Use Case 20: View owns courses

Primary Actor: Instructor

Description: Describes the process when an instructor views the course he/she is teaching.

Precondition: Instructor must have been successfully logged in.

Postcondition: If viewing course is successful, the instructor will be able to see the list of courses he/she is teaching.

Main Scenario:

1. The instructor clicks on the button to view all courses.
2. The instructor clicks on the button to filter all courses except for his own.
3. The instructor views the TA allocations of his/her course.
4. If viewing is successful, the instructor will be able to see the courses he/she is teaching and also view the allocations per course/lab.

Extensions:

1. If course data is incorrectly displayed or not displayed at all
 - Instructor can contact the admin to fix the issue.

Use Case 21: Fill out profile questions

Primary Actor: Student

Description: Describes the process when the student fills out a questionnaire in the TA-application.

Precondition: The student must have been successfully logged in.

Postcondition: The student has submitted it and can view profile answers in his or her profile.

Main Scenario:

1. The student clicks on the button to edit his profile.
2. The student sees a list of questions to fill out and fills them out.
3. The student presses submit.

Extensions:

1. The student accidentally refreshes the page or closes the tab.
- unfortunately, he or she must redo the application.

Use Case 22: Apply for TA position

Primary Actor: Student

Description: Describes the process when a student applies for a TA position.

Precondition: Student must have been successfully logged in.

Postcondition: If successful TA application, TA application-related databases are updated.

Main Scenario:

- Student clicks on the button to apply for a TA position.
- The student fills out a form. The student choose a preference for a subject. The student uploads his or her transcript. The student inputs the courses he or she has done.
- Student is prompted to verify the application details.
- If application is confirmed, student is sent to the application details page
- The application status is updated and the Application database table is updated.

Extensions:

1. The Student inputs the wrong information
- Student can view the view the application submitted and cancel the application.

Use Case 23: Check TA application status

Primary Actor: Student

Description: Describes the process when a student checks his or her application status.

Precondition: Student must have been successfully logged in.

Postcondition: The student is informed of his or her application status.

Main Scenario:

1. Student clicks on the button to see status.

2. Student views the application status.

Extensions:

1. The Student has not applied yet
 - The application informs the user no application exists and gives a button linking to the TA application page.

Use Case 24: View details of submitted application

Primary Actor: Student

Description: Describes the process of how a student can view the details of submitted application.

Precondition: Student must have submitted an application

Postcondition: The student has viewed the details of the submitted application.

Main Scenario:

1. The student logs in and goes to the application status page.
2. The student can click on a link or button to see details of application submitted.
3. On a separate page or as a dropdown, the student can see the details of the application submitted.

Extensions:

Use Case 25: Accept TA offer

Primary Actor: Student

Description: Describes the process when a student accepts an offer.

Precondition: Student must have already applied.

Postcondition: On acceptance, the TA-coordinators can now allocate the student to a particular course.

Main Scenario:

1. The student logs in and goes to application status page.
2. The student sees an offer in the status page and clicks on accept offer
3. The student verifies the acceptance after the application asks for verification.

4. The application confirms the acceptance and the database is updated for the next action to be taken by the coordinator.

Extensions:

1. The student logs in after the deadline of accepting the offer
 - The application informs the user of the unfortunate circumstance, and the student cannot accept the offer.
2. By some system error, the student accepts an offer for a course when a different TA has already been allocated to the course.
 - The application informs the user the offer has expired and provides the student with some way to contact the coordinator for any inquiries.

Use Case 26: Deny an offer

Primary Actor: Student

Description: Describes the process when a student accepts or denies an offer.

Precondition: Student must have already applied.

Postcondition: On denial,

Main Scenario:

1. The student logs in and goes to the application status page.
2. The student sees an offer in the status page and clicks on deny offer.
3. The student verifies the denial.
4. The application confirms the denial and the database is updated for the next action to be taken by the coordinator.

Extensions:

1. The student logs in after the deadline of accepting the offer
 - The TA's application is automatically denied.

Use Case 27: View how schedule looks like

Primary Actor: Student

Description: Describes the process of when a student checks his schedule

Precondition: Student must have logged in.

Postcondition: The student has seen the details of what one week looks like.

Main Scenario:

1. The student logs in and goes to the TA-schedule page.
2. The student sees a one-week spanning calendar that shows the times and days of the week during the semester that his or her courses (to TA) are in.

Extensions:

1. The student's application is not complete.
 - The TA does not see anything in the calendar.

Use Case 28: Cancel Application

Primary Actor: Student

Description: Describes the process of how a student cancels his or her own application.

Precondition: Student must have submitted an application

Postcondition: The student has canceled his or her application.

Main Scenario:

1. The student logs in and goes to the application status page.
2. The student can click on a link or button to cancel an application.
3. The application asks for verification for cancelling.
4. The student verifies and the application is canceled.

Extensions:

1. The Coordinator happens to have accepted the application and updated the offer just before the student clicks on "cancel".
 - The application redirects them to the Application status, where the new status is shown. There, the student can deny the offer.

Use Case 29: Send inquiry to coordinator

Primary Actor: Student

Description: Describes the process of how a student sends an inquiry to the coordinator

Precondition: Student is logged in.

Postcondition: The student has sent an inquiry to the coordinator. The coordinator receives an email.

Main Scenario:

1. The student clicks on a button to send an inquiry to the coordinator
2. The student fills out a textbox or form about the inquiry and submits it.
3. The system sends an email to the coordinator containing the contents of the inquiry and the student who sent it.
4. The system sends an email to the student containing the contents of the form that he sent to the coordinator.

Extensions:

Use Case 30: Indicate availability

Primary Actor: Student

Description: Describes the process of how a student indicates their time availability

Precondition: Student is filling out the application

Postcondition: The student has indicated their time availability. The coordinator can see which courses the student is taking to see their available times when allocating.

Main Scenario:

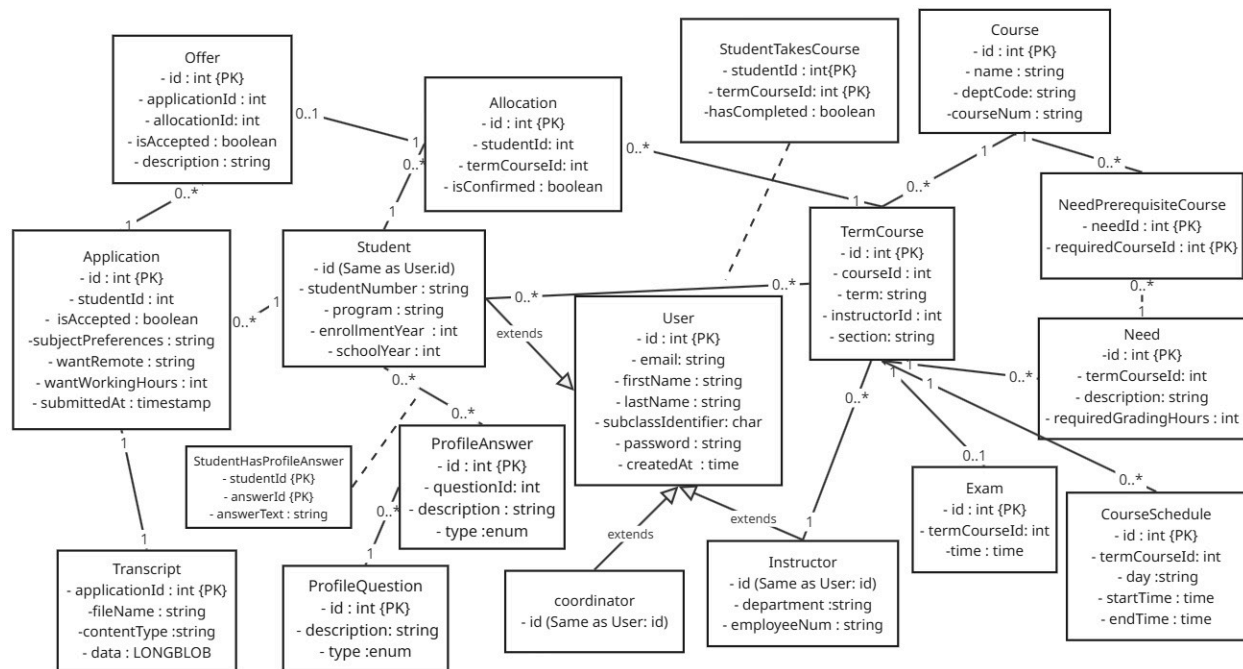
1. The student is filling out the application and indicates which courses he or she will take this semester.
2. The system saves which courses that student is taking.
3. The system shows the courses the student is taking to the coordinator when the coordinator is allocating TAs.

Extensions:

1. The student makes a mistake in their application.
 - the student can delete the application and redo it.

ER Diagram:

ER diagram



Student, Instructor, Coordinator, Course tables:

Student, Instructor, and Coordinator are the actors of this application. They all extend User because they have many fields in common with each other. Notice that their IDs are identical to the ID of the id field in User that they correlate to.

The coordinator table is the admin table. The admin is the scheduler and has all privileges for editing other information in the database.

We expect the instructor to have an employee UBC number, so Instructor.employeeNum exists.

Course.deptCode is the 4 letter identification string of a course. For example, "COSC."
Course.courseNum is the 3-digit long integer. For example, "111."

Application, Offer, Allocation tables:

Offer is connected to both Application and Allocation. This means that an offer cannot exist without the two tables. This is done because Offer can only exist if an allocation exists for the student and if the student has made an application.

Notice that Allocation can be created and connected to the student without an Offer. This is done for scalability and unforeseen circumstances such as entering a student with the allocation when the student has made no application. When the coordinator uploads the

CSV of past courses and the CSV contains past TA allocation records, the offers made for those past allocations are unnecessary for the database.

Transcript is a requirement for every application, but this may be changed as we learn more about the client's preferences.

TermCourse and Course tables:

TermCourse is a table that represents a course that lasts one semester. Course and TermCourse had to be two separate tables due to the edge case where NeedPrerequisiteCourse must refer to not a specific course taken in some term at some year, but a general course name.

TermCourse and Course allows specific recordings of which courses a student takes or has taken at some term in some year.

TermCourse.section allows the extra string you often see after a course name: "001" or "002" or "L01" or "L02." It is through the section field that a lecture separates itself from a lab.

CourseSchedule and Exam tables:

These are tables that are important for the visual scheduling application. The data in these courses will be used to create slots in a 7-day weekly scheduling application where the coordinator can compare TA times with coordinator times.

The Exam table may be deleted or kept as we learn more about the client's preferences and as the project grows. So far, exam scheduling is not expected in our project requirements, but we have added it in the table for any possible developments.

Need and NeedPrerequisite tables:

These tables record the instructor's needs for their TermCourse. The NeedPrerequisiteTable acts like an array of different courses that the instructor expects a student to have completed for a particular course. Hence, NeedPrerequisiteTable is connected to Course, as the instructor does not expect the student to have completed a course at a certain term in a certain year, but the course just generally.

ProfileAnswer, ProfileQuestion, StudentHasAnswer tables:

These tables are for the questionnaire in the TA-profile. The TA-profile will contain questions the TA will answer and they will be saved permanently and be viewable to the coordinator at any time.

ProfileQuestion.type is an enum. It can be one of the following: "TEXT," "MC_SINGLE," "MC_MULTI." As the names of the enum implies they are respectively: textbox, single-choice multiple choice, and multiple-choice multiple choice.

ProfileAnswer.type is an enum. It can be one of the following: "TEXT," "MC," "MC_TEXT". They are respectively: textbox, a checkbox, a checkbox with a text option.

StudentHasAnswer is the table that contains the answers of a student. If the answer requires text, the StudentHasAnswer.answerText gets filled. If the ProfileAnswer is a Multiple Choice question, StudentHasAnswer.answerText is null. It is interesting to note that a ProfileAnswer can be a multiple choice option and a textbox at the same time. StudentHasAnswer accommodates this by the answerText field.

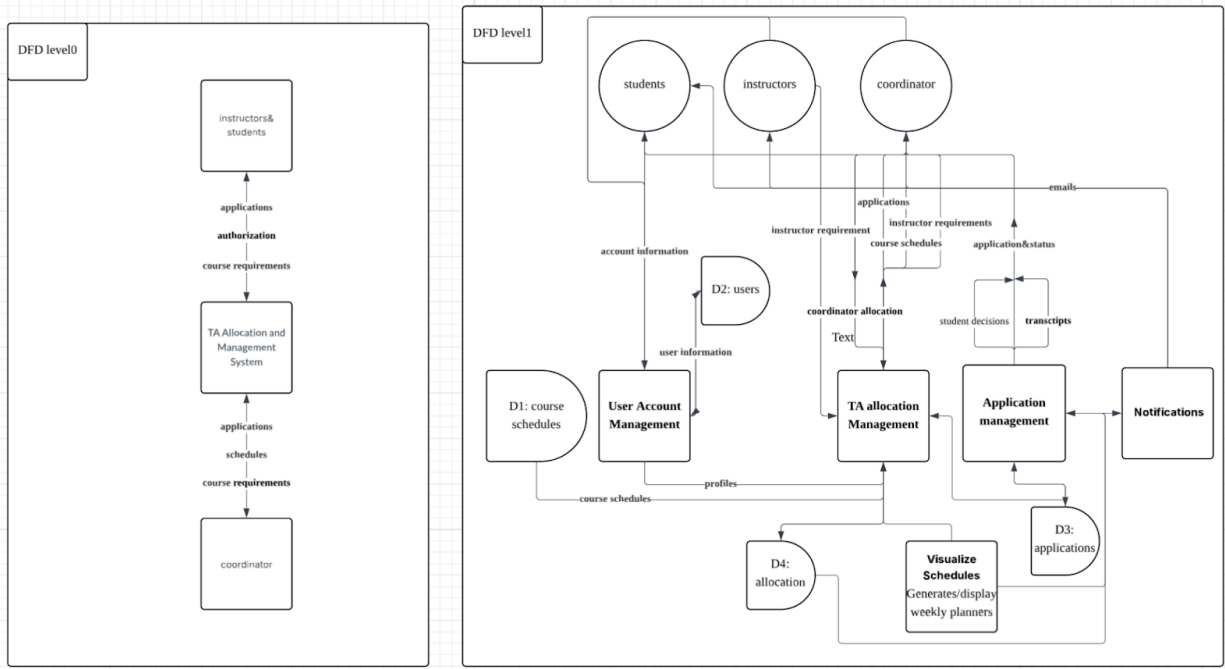
The system must satisfy the rules below by running triggers to ensure stability in the DB-level:

Rule #1: If ProfileQuestion.type = "TEXT," then ProfileAnswer.type must be of type "TEXT"

Rule #2: If ProfileQuestion.type is either "MC_SINGLE" or "MC_MULTI," then ProfileAnswer.text must be either "MC" or "MC_TEXT".

Rule #4: When a ProfileQuestion is deleted or updated, all ProfileAnswers and StudentHasAnswers referring to it should be deleted as well.

DFD



Data-Flow Diagram Description

Level 0 DFD — TA Allocation and Management System (TAMS)

The Level 0 diagram shows only the data exchanges between the system and its external parties.

External Entities

- **Instructors & Students** – grouped together at Level 0 to keep the diagram compact
- **TA Coordinator** – administrative role that oversees instructors&students's requirements and make the allocation

Data Flows

- **Course Requirements** : instructors / coordinator to the system
- **TA Applications**: from students to the system
- **Allocation Schedule**: from the system to TA Coordinator
- **Authorization Results**: from the system to instructors & students

Process-level Description

1. **Receive inputs**: The system ingests Course Requirements and TA Applications.

2. **Authenticate & validate:** Each submission is verified for identity and completeness.
3. **Match candidates to courses:** A matching engine pairs eligible students with open TA positions.
4. **Produce a draft schedule:** The resulting Allocation Schedule is sent to the TA Coordinator.
5. **Finalize & notify:** After coordinator approval, the system issues Authorization Results indicating acceptance, wait-list, or rejection.

At Level 0, the TA Allocation and Management System receives course requirements from instructors (or the coordinator) and TA applications from students.

After authenticating each request, it stores both data sets and executes a matching routine that pairs eligible students with courses. The resulting allocation is formatted into a comprehensive schedule, sent to the TA coordinator for confirmation. Once approved, the system finalises the assignments and returns authorization information to students and instructors, telling each applicant whether they have been accepted, wait-listed, or rejected.

Level 1 DFD — TA Allocation and Management System (TAMS)

External Entities

- **Students** – create accounts, submit TA applications and transcripts, and receive status notifications
- **Instructors** – define course-specific TA requirements and receive allocation results
- **TA Coordinator** – oversees the overall TA allocation process, confirms final schedules, and communicates with both students and instructors

Processes

- **User Account Management** – handles student and instructor accounts, including authentication and profile storage
- **Application Management** – receives TA applications and transcripts from students, stores them, and tracks application status
- **TA Allocation Management** – takes in course requirements and student data, then matches students to open TA positions. This process utilizes user information from D2: users and application data from D3: applications to perform allocation.
- **Visualize Schedules** – generates and displays weekly planners based on finalized allocations. This process reads allocation data from D4: allocation and course schedule data from D1: course schedules

- **Notifications** – sends emails to stakeholders regarding application decisions and allocation outcomes

Data Stores

- **D1: Course Schedules** – Stores course schedule information.
- **D2: Users** – Stores user account information and profiles for students and instructors.
- **D3: Applications** – Stores TA applications and transcripts from students.
- **D4: Allocation** – Stores finalized TA assignment schedules.

Data Flows

- **Account Information** – from students and instructors to User Account Management
- **Applications & Transcripts** – from students to Application Management
- **Instructor Requirements & Course Schedules** – from instructors and coordinator to TA Allocation Management
- **Coordinator Allocation Data** – from TA Allocation Management to the coordinator and Visualize Schedules
- **Application Status & Allocation Results** – from Notifications to students and instructors
- **Emails** – sent from Notifications to students, instructors, and coordinator
- **Course Schedules** – shared with Visualize Schedules for display
- **Profile Data (user information & profiles)** – stored in and retrieved from internal user data stores (D2: users)
- **Student Decisions** – from Application Management to D3: applications
- **Transcripts** – from Application Management to D3: applications
- **User Information** – from User Account Management to D2: users
- **Profiles** – from User Account Management to D2: users
- **Course Schedules (from D1)** – from D1: course schedules to TA Allocation Management and Visualize Schedules
- **Applications (from D3)** – from D3: applications to TA Allocation Management
- **Allocation (from D4)** – from D4: allocation to Visualize Schedules

Process-level Description

1. **Create and manage accounts:** The system receives account information from students and instructors and stores it in the user database (D2: users) via the User Account Management module.
2. **Collect applications:** Students submit TA applications and transcripts. These are stored and tracked by the Application Management module in D3: applications

3. **Match students to courses:** Course requirements and schedules are received from instructors and the coordinator.
4. **Generate draft and final schedules:** The TA Allocation Management module visualizes student schedule for the TA Coordinator, and the coordinator determines allocation. The resulting allocation data is also stored in D4: allocation. The Visualize Schedules module reads from D4: allocation and D1: course schedules data to generate and display planner views.
5. **Notify stakeholders:** Once the coordinator makes allocation, the finalized data is used by the Notifications module to send emails to students and instructors to inform them of application outcomes and final TA assignments. Students receive updates on their application status and assigned roles.

At Level 1, the TA Allocation and Management System is divided into multiple subsystems. Students and instructors first provide account information, which is stored for authentication and profile use. Students then submit TA applications and transcripts, which are managed by the Application Management module.

Instructors and the TA coordinator upload course requirements and course schedules, which are passed to the TA Allocation Management module. This module cross-references user qualifications and applications, producing an allocation schedule that is submitted to the coordinator. Once approved, the finalized data is used by the Visualize Schedules module to generate weekly planners, and by the Notifications module to send emails to students. Students and instructors receive updates on their application status and assigned roles, closing the loop in the allocation process.

UI Wireframes

1. Instructor Wireframes

Here are some of the instructor wireframes we've completed for the TA Allocation and Management System. They are designed to simplify key instructor tasks like course management, TA requirement updates, and allocations.

Navigation Guide – Instructor

The Instructor interface provides a streamlined and functional navigation layout tailored to teaching-related tasks:

- **Sidebar Navigation:**
The vertical sidebar on the left contains quick-access icons for key sections including **Home**, **My Courses**, **TA Requirements**, **TA Allocations**, **Profile**, and **Logout**. These options are clearly labeled and icon-based, allowing instructors to efficiently navigate between pages related to their courses and TA needs.
- **Home Button:**
The **Home** button leads instructors to their personalized dashboard, where they can view a summary of their courses, TA requirement statuses, and allocations. It serves as the central hub for managing their academic responsibilities.
- **Back Button:**
For pages such as TA requirement editing or allocation details, a **Back** button appears within the page content. This allows instructors to return to the previous screen without needing to rely on the sidebar, supporting smoother navigation across workflows.

This is the Instructor Login page for the UBC TA Allocation and Management System. Instructors can sign in using their email and password to access features like managing TA applications and course requirements. The page includes navigation links (Home, About, Contact), a sign-up option, and a “Forgot password?” link for account recovery. The header and footer maintain UBC branding, with useful links like Terms of Use, Privacy, Accessibility, and Help. The background image adds visual appeal while the clean layout ensures ease of use.

Login Page

**UNIVERSITY OF BRITISH COLUMBIA**

[Home](#) [About](#) [Contact](#) [Sign in](#) [Sign Up](#)

Welcome to TA Allocation and Management System

Username/ email

Password

[Forgot password?](#)

[Login](#)



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Accessibility
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Next, is the **Sign-Up** page. It allows new users—such as instructors, students, or coordinators—to create an account by entering their full name, email, phone number, role, and a secure password. The form includes password confirmation to ensure accuracy. Navigation links (Home, About, Contact) are available at the top, while existing users can click "Login Here" to return to the sign-in page.

Sign up

 UNIVERSITY OF BRITISH COLUMBIA  TA Portal

[Home](#) [About](#) [Contact](#) [Sign in](#)


TA Portal
Efficient TA Allocation Made Easy

Create an Account

Full Name*

Email*

Phone Number*

Role* ▼

Password*

Confirm Password*

Already have an account?
[Login Here](#)

Signup

 [Terms of Use](#) [Privacy](#) [Copyright](#) [Accessibility](#) [Need Help?](#) 

Next is, **Instructor Homepage** displaying a personalized summary for the logged-in instructor. It lists all assigned courses along with their TA requirement and allocation status. Each course card clearly shows whether the instructor has completed the requirement details and whether any TAs have been assigned. "View Details" links offer quick access to update or review each section. The sidebar provides intuitive navigation for core tasks like viewing courses, submitting TA requirements, checking allocations, and managing the profile. This page focuses on streamlining the instructor's workflow by highlighting pending actions at a glance

Instructor homepage



UNIVERSITY OF BRITISH COLUMBIA



TA Portal



Home



My Courses



TA Requirements



TA Allocations



Profile



Logout

Welcome Dr. Ched,

Summary

COSC 111

Introduction to Computer Science

Allocated TA(s)
Not Assigned

[View Details](#)

Requirement(s)

Not Completed

[View Details](#)

COSC 111

Introduction to Computer Science

Allocated TA(s)
Not Assigned

[View Details](#)

Requirement(s)

Not Completed

[View Details](#)

COSC 111

Introduction to Computer Science

Allocated TA(s)
Not Assigned

[View Details](#)

Requirement(s)

Not Completed

[View Details](#)



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Another one is, **Requirement(s) Update** page allows instructors to define or edit TA prerequisites and grading hours for their courses. Instructors can search and filter by term or course, view existing details, and click “Update” to edit them. The bottom section enables input of specific course prerequisites (e.g., COSC 011) and estimated grading hours, which can be submitted to update the system. This page ensures TA requirements are clear and up to date for proper TA allocation.

Requirement(s) update



UNIVERSITY OF BRITISH COLUMBIA





Home



My Courses



TA Requirements



TA Allocations



Profile



Logout

Requirement(s) Update

Winter 2025

All Courses

COSC 111 Introduction to Computer Science

Monday- Wednesday11.00 - 12.30pm

Grading hours: None

Prerequisites: None

CancelUpdate

COSC 111 Introduction to Computer Science

Prerequisites

COSC 011 X

Grading Hours

Submit

[← Back](#)



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Next is, **TA Allocations** page displays a list of courses with their assigned teaching assistants and the number of hours each TA is scheduled. Instructors can filter results by term or course, and search using the top bar. Each course entry shows its schedule and the names and hours of allocated TAs. At the bottom, there's an option to export the allocation data in CSV format for record-keeping or reporting. This page helps instructors keep track of current TA assignments in a clear and organized manner.

View TA Allocation page

**UNIVERSITY OF BRITISH COLUMBIA**


Home


My Courses


TA Requirements


TA Allocations


Profile


Logout

TA Allocations

Winter 2025 ▾

All Courses ▾

COSC 111 Introduction to Computer Science
Monday- Wednesday
11.00 - 12.30pm

 Ron Fisher 8 hours
 Will Smith 10 hours

COSC 222 Introduction to Data Structure
Tuesday- Thursday
9.00 - 11.30pm

 John Lee 8 hours
 William R 10 hours

[← Back](#)[CSV ▾](#)[Export](#)



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This **Instructor Profile** page allows users to view and update their account information. It displays fields such as User ID, Department, Employee ID, Name, Email, Phone, and Role. Instructors can also change their password and confirm the update. The interface includes options to edit or delete the profile, along with a "Save Changes" button to apply updates. This page helps ensure instructors can manage their personal and professional details.

Instructor Profile



UNIVERSITY OF BRITISH COLUMBIA



TA Portal



Home



My Courses



TA Requirements



TA Allocations



Profile



Logout

User Profile

User ID: Department :

Employee ID :

Personal details

Name : Email :

Phone : Role :

Change Password:

Confirm Password:

Edit Profile

Delete Profile

Save Changes

[← Back](#)



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Need Help?



2. TA coordinator wireframes

Below are some of the TA Coordinator wireframes we've completed for the TA Allocation and Management System. They are designed to streamline core coordinator tasks such as managing course data, reviewing TA applications, allocating TAs, and overseeing profile information. The sidebar provides quick access to each function, ensuring smooth navigation and an efficient workflow.

Navigation Guide – TA Coordinator

The TA Coordinator Portal includes a consistent and intuitive navigation structure to manage key responsibilities efficiently:

- **Sidebar Navigation:**
Located on the left side of every screen, the sidebar offers quick access to essential pages such as **Home**, **Courses Management**, **TA Applications**, **TA Allocations**, **Profile**, and **Logout**. Each item is represented with a clear icon and label, making it easy to switch between different coordination tasks.
- **Home Button:**
The **Home** button takes coordinators back to their main dashboard, which summarizes key information like pending TA applications, course needs, and recent activity. It serves as the central starting point for all coordinator functions.
- **Back Button:**
On detailed pages like course editing or application review, a **Back** button is provided in the main content area (bottom-left). It allows coordinators to return to the previous screen quickly without needing to navigate through the sidebar again.

The TA Coordinator uses the same **login and signup pages** as the instructor, providing a unified and consistent entry point into the system. While the interface remains the same, the system identifies the user's role after login and displays role-specific features accordingly. This approach simplifies the user experience while maintaining secure and tailored access for each role.

Below is **TA Coordinator Dashboard** provides an overview of key metrics and recent actions. At a glance, the coordinator can see the number of pending TA applications, courses needing TAs, and offers awaiting acceptance. A dropdown allows filtering by term. The Quick Actions section lists recent TA applications with applicant details, course assignments, status updates, and options to review or view each application. The left sidebar offers navigation to core pages like Courses Management, TA Applications, TA Allocations, Profile, and Logout.

TA co-ordinator Homepage



UNIVERSITY OF BRITISH COLUMBIA





Home



Courses Management



TA Applications



TA Allocations



Profile



Logout

Coordinator Dashboard

Welcome Dr. John

21

Pending TA Applications

10

Courses Requiring TA's

5

Offer sent waiting on acceptance

Winter 2025 ▾

Quick Actions

Recent TA applications

Applicant	Course	Status	Action
John D. Mathews	N/A	Pending	Review
Mathews	COSC 222	Accepted	View
Mark Devis	COSC 221	Rejected	View
Will Jack	COSC 310	Accepted	View



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This **Course Management** page allows the TA Coordinator to add new courses to the system. Courses can be uploaded in bulk using a .csv file via drag-and-drop or file selection. The uploaded data appears in a table for review before approval. Alternatively, courses can be added manually by filling out details such as course name, department number, course number, section, term, instructor ID, and prerequisites.

Course Management Page

UNIVERSITY OF BRITISH COLUMBIA

TA Portal

Home

Courses Management

TA Applications

TA Allocations

Profile

Logout

Add Course(s)



Drag & drop your .csv file here
or select from your computer

CancelConfirm upload

Course Name	Dept Num	Course Num	Section	Instructor ID	Term	Prerequisites

Approve

Or Manually

Course Name:

Dept Num: Course Num:

Section: Term:

Instructor ID

Prerequisite(s)

CancelAdd Course

← Back



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Need Help?



This **TA Applications** page enables the TA Coordinator to review and manage applications submitted by students. Coordinators can filter applications by term, status, or course, and view a summary list showing each applicant's course preferences, status, and requested hours. Clicking "View" reveals detailed information such as availability, completed courses, and preferred allocations. Coordinators can then assign the applicant to a course or update their application status with options to approve or reject.

TA Applications



UNIVERSITY OF BRITISH COLUMBIA



TA Portal



Home



Courses Management



TA Applications



TA Allocations



Profile



Logout

Term ▾

Status ▾

Search

Course ▾

Applicant	Course Preferences	Status	Hours Requested	Action
John D.	COSC 111	Accepted ▾	18	Review
Mathews	COSC 222	Pending ▾	18	View
Mark Devis	COSC 221	Accepted ▾	18	View
Will Jack	COSC 310	Rejected ▾	18	View

Applicant: Mathews

Course Preference : COSC 222, COSC 221

Availability: All days except Fri

Hours Requested: 20

Course Completed: COSC 111, COSC 121

Allocate To

COSC 221 ▾

 Mon Fri 9.00 - 11.30

Cancel

Reject

Approve

[← Back](#)



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Need Help?



This **TA Allocations** page allows the TA Coordinator to match courses with suitable applicants based on filters and availability. Coordinators can filter courses by department, section, and term, and filter applicants by their preferences, availability, and requested hours. The page displays detailed information about the selected course and applicant, including prerequisites, schedule, and grading hours. A visual **weekly calendar** compares course slots with TA availability, helping coordinators quickly identify scheduling conflicts or suitable matches.

TA Allocations



UNIVERSITY OF BRITISH COLUMBIA



TA Portal

Home

Courses Management

TA Applications

TA Allocations

Profile

Logout

TA Allocations

Course Filter

Dept Code

Section

Term

Filter

Applicant Filter

1st Preference

Term

2nd Preference

Hours

2nd Preference

Search

Results

COSC 111

COSC 222

COSC 121

John D.

Alice lee

Kyle J.

COSC 222 Data Structure

Pre-requisite(s): COSC 121, COSC 111

Schedule: Mon Fri 9.00 - 11.30

Grading hours Needed: 12

Kyle Jamieson

Completed Course(s): COSC 111

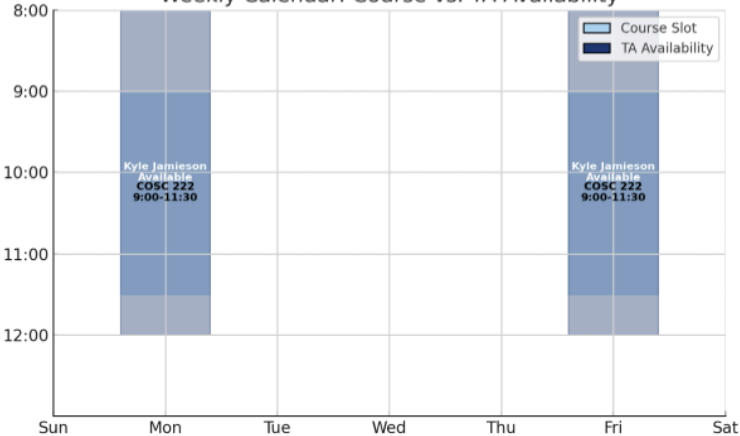
Availability: Mon Fri

Grading hours requested: 20

Pre-requisite Status: Incomplete

Grading hours : Met

Weekly Calendar: Course vs. TA Availability





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Need Help?



3. TA wireframes

Below are some of the TA wireframes we've completed for the TA Allocation and Management System. They are designed to support teaching assistants in viewing assigned courses, submitting applications, tracking application status, and managing personal profiles. The sidebar provides intuitive access to each feature, making the overall experience simple and efficient for TAs.

Navigation Guide

The TA Portal uses a clear and structured navigation system designed for ease of use:

- **Sidebar Navigation:**
Located on the left side of every page, the sidebar provides quick access to key sections such as Home, My Courses, Submit Application, My Applications, Profile, and Logout. Each icon is labeled and consistently styled, allowing users to move between core functionalities without confusion.
- **Home Button:**
The Home button at the top of the sidebar brings users back to their dashboard or landing page based on their role (e.g., TA, Instructor, Coordinator). It acts as a central point to reset navigation and return to a familiar overview.
- **Back Button:**
Some pages (like detailed views or submission forms) include a Back button bottom-left corner of the main content area. This button allows users to return to the previous screen without needing to reselect options from the sidebar.

The TA uses the same login and signup pages as the instructor, providing a unified and consistent entry point into the system. While the interface remains the same, the system identifies the user's role after login and displays role-specific features accordingly. This approach simplifies the user experience while maintaining secure and tailored access for each role.

This **TA Homepage** provides a personalized summary for teaching assistants. Upon logging in, users can view the status of their TA applications for each course, such as "Not Completed," "Completed," or "Accepted." Each course card includes a "View Details" link for further action or updates. The left sidebar allows navigation to key sections like My Courses, My Applications, Profile, and Logout.

TA Homepage



UNIVERSITY OF BRITISH COLUMBIA



TA Portal



Home



My Courses



My Applications



Profile



Logout

Welcome Jeremy

Summary

COSC 111

Introduction to Computer Science

Application Status
Not Completed [View Details](#)

COSC 121

Computer programming II

Application Status
Completed [View Details](#)

COSC 222

Data structure

Application Status
Accepted [View Details](#)



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The **My Courses** page displays a TA's assigned courses for the selected term. Each card shows course name, schedule, assigned grading hours, and TA status. A term filter and export option are available for easy access and record-keeping. This page helps TAs quickly view and manage their teaching assignments.

My Courses



UNIVERSITY OF BRITISH COLUMBIA



TA Portal



Home



My Courses



My Applications



Profile



Logout

My Courses

Winter 2025

Export

COSC 111
Introduction to Computer Science
Tues & Thurs 9.00-11.30AM
Instructor
Assigned Grading Hours 10
TA Status - Active

COSC 121
Computer Programming II
Mon & Wed 11:00-12.30AM
Instructor
Assigned Grading Hours 12
TA Status - Active

COSC 222
Data Structure
Tues & Thurs 9.00-11.30AM
Instructor
Assigned Grading Hours 10
TA Status - Active



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This **Application Submission** page allows TAs to apply for a position by filling out essential details. It includes dropdowns for course preferences (1st, 2nd, 3rd), a field for requested working hours, and a weekly availability selector with time slots. TAs can list completed courses, upload their transcript, and indicate whether they prefer remote work. A clear "Submit" button finalizes the application, while "Cancel" lets users exit the form.

Application submission

 UNIVERSITY OF BRITISH COLUMBIA 

 Home
 My Courses
 Submit Application
 My Applications
 Profile
 Logout

Application Submission

Course preference(s)*

1st Preferences ▾ 2nd Preferences ▾ 3rd Preferences ▾

Hours Requested*

0 ▾

Availability*

☐ S ☐ M ☐ T ☐ W ☐ T ☐ F ☐ S

Start Time

1:30 PM

End Time

3:30 PM

Completed Course(s)*

COSC 111 X

Transcript*

No file Chosen

Want remote work*

☐ Yes ☐ No



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